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Yitran Technologies Ltd.

Powerline Communication (PLC)

What is Power Line Communication (PLC)?

•**Power Line Communication (PLC)** is a technology that enables **data communication over existing electrical power lines**.

•It allows the **electric grid itself to become a communication network**, eliminating the need for separate communication infrastructure.

•PLC is widely used in:

- **Smart Electric Metering (AMI/AMR)**
- **Energy Management Systems**
- **Home Automation and Control**
- **Smart Street Lighting**
- **Internet of Things (IoT) applications in utilities**

PLC is Part of a Hybrid Communication Architecture

•**No single communication technology fits all deployment conditions.**

•Smart metering systems typically use a **hybrid communication architecture**

- **PLC (Power Line Communication)**
- **RF Mesh**
- **Cellular (GPRS / 4G / LTE)**

- **PLC is especially useful when:**

RF communication is unreliable (dense buildings, basements, underground meters, heavy RF noise).

•**Large meter deployments make cellular fees expensive**.

•Utilities prefer **owning and controlling the communication infrastructure** rather than relying on telecom operators.

Why PLC is Particularly Suitable for India's Smart Meter Rollout?

•In India, **PLC is not only an alternative communication technology** — it also **adds important grid intelligence capabilities** that other communication technologies cannot easily provide.

•Because PLC communicates **through the electrical network itself**, it enables utilities to obtain **valuable information about the grid topology and operating conditions**.

Key capabilities enabled by PLC include:

- **Automatic Meter Indexing and Tagging**
  - Identifies which meters are connected to which transformer and feeder.
- **Phase Identification**
  - Determines the exact phase (R/Y/B) each meter is connected to.
- **Grid Topology Discovery**
  - Automatically maps how meters are connected across the distribution network.
- **Line Condition Monitoring**
  - Detects abnormalities or changes in line conditions.
- **Transformer Area Mapping**
  - Associates each meter with its serving transformer.
- **Support for Loss Reduction Programs**
  - Helps utilities detect **energy theft, technical losses, and abnormal connections**.

Cost Impact of Communication Technology In Smart Meter Program

India is planning one of the **largest smart meter deployments in the world**, with **hundreds of millions of meters**.

The **communication technology choice has a major impact on total system cost**. **Usually, the communication module is the most expensive part by far.**

Cellular communication (GPRS / LTE) usually involves:

- **Recurring monthly fees per meter**
- Dependence on **telecom operators**
- Long-term **operational expenditure (OPEX)**

When deployed across **tens or hundreds of millions of meters**, these fees can become **a very significant national cost**.

PLC eliminates the need for per-meter telecom subscriptions, since communication occurs **over the existing power grid**.
This allows utilities to:

- **Reduce long-term operating costs**
- **Maintain full control of the communication infrastructure**
- **Scale deployments more economically**

Cost Impact of Communication Technology In Smart Meter Program

Communication Type	Infrastructure Needed	Recurring Fees	Utility Control
Cellular (GPRS LTE)	Telecom network	Yes-per meter	Low
RF Mesh	RF network infrastructure	Limited	Medium
PLC	Existing electric grid	**No per-meter fee**	High

PLC Enables Better Grid Management

PLC not only provides communication for smart meters — it also **improves the operational management of the distribution grid**.

Operational benefits include:

- **Improved load balancing across phases**
  - Phase identification helps utilities **redistribute loads and reduce transformer stress**.
- **Faster fault detection and troubleshooting**
  - Utilities can more easily **identify where communication or power interruptions occur**.
- **Better transformer load monitoring**
  - PLC networks allow utilities to monitor **how many meters and how much load are connected to each transformer**.
- **Improved planning for network upgrades**
  - Accurate grid connectivity information helps utilities **plan feeder upgrades and capacity expansion**.
- **Better outage localization**
  - Utilities can quickly determine **which meters are affected by outages** and **narrow down fault locations**.

Why PLC Works Well in Dense Urban Environments

•In many **urban and semi-urban areas**, RF communication face challenges due to:

- **Dense building structures**
- **Basements and indoor meter installations**
- **Concrete and metal barriers**
- **RF interference and spectrum congestion**

•**Power Line Communication avoids these obstacles** because communication signals travel **through the electrical wiring instead of through the air**.

•This makes PLC particularly effective in environments where:

- **Meters are installed inside buildings**
- **RF signal penetration is limited**
- **Urban density causes RF interference**

•PLC enables **reliable communication between meters and the distribution transformer**, even in **complex urban electrical networks**.

•As a result, PLC can **increase communication reliability and coverage** in areas where RF networks may struggle.

"PLC Will Not Work on the Indian Grid"… or Will It?

Key Points

•A common perception is that **PLC will not work reliably on the Indian electricity distribution grid**.

•The concerns usually relate to:

- **High electrical noise**
- **Long feeder lines**
- **Complex distribution topology**
- **Frequent network changes**

•However, **not all PLC technologies are the same**.

•**Yitran PLC technology was specifically designed to operate in difficult power line environments**, including:

- **High noise conditions**
- **Low-quality power lines**
- **Long-distance distribution feeders**

•Key characteristics of **Yitran PLC** include:

- **Robust modulation designed for noisy power lines**
- **Adaptive communication techniques**
- **High immunity to electrical interference**
- **Proven operation in challenging grid environments**

•As a result, **PLC can work reliably even on complex and noisy grids such as**

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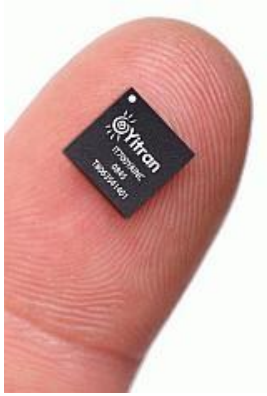

THANK YOU